

Srikrishnaa Vadivel

Machine Learning Engineer | Scientific ML | Equivariant Models

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Summary

PhD researcher applying machine learning to first-principles materials and molecular systems. Experience spans PyTorch, equivariant graph networks, diffusion models, distributed training concepts, HPC data pipelines, and physics-based loss design.

ML Experience

Yale University, PhD Researcher

2021–present

- Developed first-principles datasets and workflows for self-trapped exciton calculations on leadership-class HPC resources.
- Prototyped EGNN-style force models with energy gradients $F_{ai} = -\partial E / \partial R_{ai}$ and molecular graph construction.
- Built diffusion-model notes and code scaffolds covering DDPM training, conditional inference, transformers, and index-notation derivations.

Probe Technologies / Weatherford

2019–2021

- Built GPU-accelerated processing and visualization for large acoustic waveform data streams using CUDA and OpenGL.
- Engineered production data tooling around ASP services and Microsoft database systems.

Selected Projects

- **EGNN QM9 Force Model:** PyTorch message-passing model, force loss, synthetic graph fallback, and training-curve generation.
- **Diffusion Notes:** DDPM equations with α_t, β_t , conditional training, and inference in index notation.
- **LiF Band Scaffold:** reciprocal-space Hamiltonian assembly and eigenvalue plotting for physics-informed ML context.

Education

Yale University, PhD Electrical Engineering, 2021–present

Yale University, MPhil Electrical Engineering, 2023

Yale University, MS Electrical Engineering, 2023

Cornell University, MEng ECE, 2017–2018

Cornell University, BA ECE, 2013–2017

Skills

PyTorch, NumPy, SciPy, pandas, scikit-learn, Hugging Face Transformers, Accelerate, datasets, CUDA, MPI, OpenMP, Linux, Python, C++, Docker, Kubernetes, PETSc, SLEPc.